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ORIGINAL ARTICLE

Comparative efficacy and safety of uricosuric agents in the treatment of gout or hyperuricemia: a systematic review and network meta-analysis

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Abstract

Objectives The current world witnesses a greatly increased prevalence and incidence of hyperuricemia and gout with unfortunately the comparative efficacy and safety of present available uricosuric agents remaining uncertain. We herein aimed to investigate the most appropriate uricosuric agent for gout or hyperuricemia patients.

Method PubMed, Embase, Cochrane Library databases, and ClinicalTrials.gov from inception to 2 July 2022 were searched to retrieve eligible studies assessing efficacy and safety of uricosuric drugs in hyperuricemia or gout patients. Network meta-analysis was carried out using the Stata 16.0 software.

Results Twelve randomized controlled trials comprising 1851 patients were eventually included. Network meta-analysis showed that dotinurad 4 mg once daily, verinurad, dotinurad 2 mg once daily, dotinurad 1 mg once daily, and benzbromarone were the top 5 effective treatments to achieve target serum uric acid. Furthermore, dotinurad 4 mg once daily was more effective at achieving urate-lowering targets (RR of dotinurad 4 mg once daily vs. probenecid: 1.68, 95% CI [1.13; 2.50]) and safer (RR of probenecid vs. dotinurad 4 mg once daily: 1.77, 95% CI [0.69; 4.56]) than probenecid.

Conclusions This network meta-analysis demonstrated an important absolute benefit of dotinurad 4 mg once daily to achieve target serum uric acid and low risk of adverse events for drug treatment of gout or hyperuricemia patients. Additionally, verinurad might be used as an alternative uricosuric therapeutic option to dotinurad. These findings provided further comprehensive insight into the treatment value of current uricosuric agents for gout or hyperuricemia.

Key Points

1. *This is the first systematic review and network meta-analysis examining the efficacy and safety of currently available uricosuric agents in gout or hyperuricemia patients.*
2. *Recommended doses of dotinurad 4mg once daily used for the treatment of gout or hyperuricemia patients can significantly decrease serum uric acid levels.*
3. *The present findings will provide further comprehensive insight into the treatment value of certain uricosuric agents for gout or hyperuricemia.*

Keywords Gout · Hyperuricemia · Network meta-analysis · Uricosuric

Introduction

Gout is the most prevalent inflammatory arthritis worldwide. The prevalence of gout ranged from 1 to 4% across the globe and its incidence increased every year [1]. Meanwhile, the prevalence of hyperuricemia, the main pathological condition in the development of gout, is also rising mainly due to the aging population, comorbidities, and other lifestyle factors [2]. In view of the fact that gout

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and hyperuricemia may increase the risk of cardiovascular disease and chronic kidney disease, they have become one of the most important cause factors of reduced life quality and increased care cost as well [2–4]. Hence, urate-lowering therapy (ULT) is urgently needed to prevent gout attacks in clinical practice.

Pharmacotherapy of ULT can be classified into xanthine oxidase inhibitors (XOIs) and uricosurics based on pharmacologic mechanism [5]. XOIs such as allopurinol or febuxostat are effective for reducing the production of serum uric acid (sUA), but they are also associated with considerable side effects, which leads to much debate about their safety [6]. Therefore, as an alternative to XOIs, uricosurics have drawn increasing attention from all over the world. Compared with XOIs, uricosurics are also highly effective in lowering sUA levels although they are often used as second-line urate-lowering agents. Benzbromarone and probenecid, the representative drugs of uricosurics, were reported to be potent in enhancing renal excretion of sUA [7]. Other uricosuric drugs such as verinurad, SHR4640, and dotinurad are undergoing clinical trials that might play important roles in the development of ULT [8]. Therefore, appropriate clinical use of uricosurics would be essential for the optimal treatment of hyperuricemia and gout.

However, although previous studies regarding uricosuric drugs demonstrated their efficacy and safety in urate-lowering therapy, certain major limitations have restricted their extensive clinical application. As a matter of fact, previous studies focused more on the general population of patients with gout or hyperuricemia with few attentions being paid to special populations such as XOI intolerance or allergy patients, which was unable to provide recommendations on how to select ULT agents for these patients [9, 10]. Furthermore, uricosurics in combination with XOIs are increasingly concerned in recent years, which was proved to be efficacious in lowering sUA levels. But the combination therapy approach is also not applicable to XOI intolerance or allergy patients [11, 12]. In addition, previous pairwise meta-analyses evaluated single ULT agents in comparison with standard care [13–18]. Nevertheless, clear research evidence to simultaneously estimate relative efficacy and safety among currently available uricosuric agents is still missing, resulting in little information on a comprehensive ranking of these agents for the treatment of gout or hyperuricemia [19].

Based on this background, we herein conducted a network meta-analysis that investigated the most appropriate uricosuric agent with high benefits (lowering sUA levels) and low risk of adverse events (AEs) for the abovementioned special populations, hoping to achieve rational and individualized use of uricosurics in the clinical treatment.

Materials and methods

Search strategy

The PubMed, Embase, Cochrane Library databases, and ClinicalTrials.gov were searched from their inception to 2 July 2022. According to the search principle of PICOS (participants, interventions, comparisons, outcomes, and study design), we conducted a search strategy that combined the free text and medical subject heading terms of the following keywords: "hyperuricemia," "gout," "uricosuric" (Supplementary Table S1). We also checked reference lists of relevant reviews and eligible studies to ensure a comprehensive search. There was no restriction of language or year of publication.

Inclusion criteria and exclusion criteria

Studies meeting the following criteria were included: (a) participants (≥ 18 years) were diagnosed with hyperuricemia with or without gout; (b) intervention group taken one of the seven uricosuric agents (probenecid, benzbromarone, lesinurad, dotinurad, verinurad, arhalofenate, or SHR4640); (c) comparison group taken placebo or another uricosuric agent different from the treatment group; (d) outcomes of interest were reported; and (e) study design was randomized controlled trial (RCT), cohort study, or case–control study. The exclusion criteria were as follows: (a) studies that combined uricosuric agents with XOIs; (b) studies that included pregnant woman, lactating women, or patients with severe comorbid conditions; (c) studies that without a designated intervention or comparator arm; and (d) not human research.

Data extraction and quality assessment

Two investigators (L.R.C. and Y.J.L.) independently extracted all data according to a data extraction form. The extracted information mainly included the first author's name, the publication year, basic characteristics of patients (gender, age, and baseline serum concentration of uric acid), treatment protocol, treatment duration (months), and study design. In the present study, the primary efficacy outcome was defined as the percentage of patients achieving serum concentration of uric acid ≤ 6.0 mg/dL at the final visit. The primary safety outcome was defined as the incidence of any AEs during the period of the trial (including abnormal liver function, renal impairment, rash, gastrointestinal disorders, joint-related signs). Besides, the secondary outcomes were the incidence of gouty arthritis and gouty flares.

After extracting the data, the methodological quality was assessed by using the Cochrane bias risk assessment

tool for included studies. The certainty of network evidence was graded by using the Grade of Recommendations Assessment, Development, and Evaluation (GRADE) for primary outcomes. Any disagreement between the two investigators was resolved by discussion and consensus. This systematic review and network meta-analysis is reported following the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) extension statement for network meta-analysis [20].

Statistical analyses

We performed random-effects pairwise meta-analyses and a network meta-analysis in a frequentist framework to obtain network estimates for the aforementioned outcomes. Before running the network meta-analysis, we first assessed the transitivity assumption to explore whether basic characteristics were sufficiently similar between comparisons by applying narrow inclusion criteria. Secondly, we evaluated the heterogeneity by conducting a pairwise meta-analysis to generate direct estimates for primary outcomes. Finally, we evaluated inconsistency assumption (the statistical disagreement between direct and indirect evidence in a closed loop) locally using a loop specific approach, and globally using a design by treatment interaction model. After the three assumptions above were assessed, we used surface under the cumulative ranking (SUCRA) to rank the efficacy and safety of intervention in the network meta-analysis. We also performed sensitivity analyses to assess the robustness of the efficacy and safety outcomes by excluding studies at high risk of bias. In addition, we compared the results of the base case analysis with "all missing failure" analysis to explore the impact of missing data; this analysis assumed that all missing patients had a negative event. To explore potential sources of publication bias, we assessed the presence of small-trial effects for primary outcomes with the comparison adjusted funnel plot.

We calculated risk ratios (RRs) for binary outcomes and presented with 95% confidence intervals (95% CIs) in the pairwise meta-analysis and network meta-analysis. All analyses were done in Stata 16.0 software using the metan and network packages for the pairwise and network meta-analysis, respectively.

Ethical approval and informed consent

All results and analyses were based on previous ethically approved studies; thus, for this type of study, formal consent is not required.

Results

Characteristics of eligible studies

Our systematic search identified 1321 citations published between 1979 and 2022. Finally, we included 12 studies [21–32] that related to 12 RCTs in the network meta-analysis. The selection process and exclusion reasons are shown in Supplementary Figure S1, and the characteristics of included studies are presented in Table 1. The publications were from 2008 to 2022, and 58.33% ($n = 7$) of studies were conducted in Asia. The study sample sizes ranged from 14 to 319 patients, while patient age ranged from 26 to 75 years, 96.87% were male. The duration of follow-up ranged from 1 to 8.5 months. Distribution of baseline characteristics was generally balanced in the included studies. In the present study, there was one five-arm study that compared dotinurad (0.5/1/2/4 mg once daily) with placebo [22]. Furthermore, there were 1 four-arm study [23], 3 three-arm studies [27, 30, 31], and 7 dual-arm studies [21, 24–26, 28, 29, 32].

The risks of bias assessments for included studies are presented in Supplementary Figure S2. Six studies had a low overall risk of bias [22, 24, 27–29, 31]. Five studies had a moderate overall risk owing to other bias [21, 23, 26, 30, 32]. One study had a high overall risk primarily because of failure to completely blind participants and personnel [25].

Primary efficacy outcome

Nine studies [21–26, 29–31] comprising 1557 patients evaluated the primary efficacy outcome. The network plot included 12 individual nodes (Supplementary Figure S3). Each of the nodes represents different active interventions or placebo; placebo was the most well-connected intervention with all other interventions. The network meta-analysis results showed that most of the active interventions were more efficacious than placebo to achieve serum concentration of uric acid ≤ 6.0 mg/dL, except for arhalofenate 600 mg daily, and arhalofenate 800 mg daily (Fig. 1). In addition, SHR4640 showed better efficacy than dotinurad 0.5 mg once daily (RR of SHR4640 vs. dotinurad 0.5 mg once daily: 2.40, 95% CI 1.18–4.90). Dotinurad 1 mg once daily showed better efficacy than dotinurad 0.5 mg once daily (RR of dotinurad 1 mg once daily vs. dotinurad 0.5 mg once daily: 2.88, 95% CI 1.58–5.24). Probenecid, benzbromarone, dotinurad 2 mg once daily, and dotinurad 4 mg once daily showed better efficacy than dotinurad 0.5 mg once daily (RR of probenecid vs. dotinurad 0.5 mg once daily: 2.37, 95% CI 1.18–4.77; RR of benzbromarone vs. dotinurad 0.5 mg

Table 1 Characteristics of the included studies

Study, year	Treatment duration (months)	*N*	Intervention	Dose	Age (m ± SD)	Male (%)	Baseline sUA (mg/dL)
M K Reinders, 2008	2	35	Probenecid	2 g	58 ± 12	94	9.24 ± 1.51
		27	Benzbromarone	200 mg	55 ± 16	100	9.07 ± 1.18
Ogino K, 2009	2	7	Placebo	/	60 ± 5	71.4	10.2 ± 0.3
		7	Benzbromarone	50 mg			
NCT01399008, 2012	1	33	Placebo	/	50.4 ± 11.0	97	≥ 6.5
		35	Arhalofenate	400 mg	52 ± 10.9	100	
		32	Arhalofenate	600 mg	50.6 ± 10.2	100	
Poiley J, 2016	3	28	Placebo	/	49.6 ± 10.0	92.9	9.1 ± 1.4
		53	Arhalofenate	600 mg	53.0 ± 11.8	96.2	9.0 ± 1.6
		51	Arhalofenate	800 mg	52.5 ± 9.8	92.2	9.3 ± 1.4
Tausche AK, 2017	6	107	Placebo	/	55.3 ± 12.0	90.7	9.18 ± 1.51
		107	Lesinurad	400 mg	53.6 ± 12.5	91.6	9.48 ± 1.50
1. Fitz-Patrick D, 2018	6	42	Placebo	/	55.0 ± 9.53	92.9	8.73 ± 0.93
		42	Verinurad	5 mg	53.3 ± 9.44	93	8.73 ± 0.93
		43	Verinurad	10 mg			
		44	Verinurad	12.5 mg			
2. Fitz-Patrick D, 2018	6	40	Placebo	/	54.5 ± 10.3	97.5	8.59 ± 0.98
		41	Verinurad	7.5 mg	51.3 ± 9.3	100	8.43 ± 0.99
		41	Verinurad	12.5 mg	52.7 ± 10.0	97.6	
		41	Verinurad	15 mg	52.3 ± 10.7	95.1	
1. Hosoya T, 2019	2	20	Placebo	/	48.8 ± 9.7	100	8.85 ± 1.30
		20	Dotinurad	1 mg	52.7 ± 9.4	100	8.79 ± 1.00
		19	Dotinurad	2 mg	50.1 ± 8.3	100	8.91 ± 1.23
		21	Dotinurad	4 mg	51.2 ± 7.4	100	8.95 ± 1.48
2. Hosoya T, 2019	3	39	Placebo	/	52.8 ± 11.0	97.44	8.93 ± 1.04
		39	Dotinurad	0.5 mg	58.5 ± 12.1	100	9.02 ± 1.20
		42	Dotinurad	1 mg	57.4 ± 12.4	100	8.84 ± 1.07
		39	Dotinurad	2 mg	55.0 ± 13.5	97.44	8.96 ± 1.14
		40	Dotinurad	4 mg	58.0 ± 10.5	97.5	8.94 ± 1.10
3. Hosoya T, 2019	8.5	276	Dotinurad	2 mg	54.2 ± 10.4	99.3	8.63 ± 1.03
		43	Dotinurad	4 mg	51.3 ± 10.5	100	9.76 ± 1.25
4. Hosoya T, 2020	3.5	98	Benzbromarone	50 mg	54.7 ± 10.4	100	8.92 ± 1.28
		102	Dotinurad	2 mg	55.0 ± 10.3	98	8.90 ± 1.16
Lin Y, 2021	1.25	38	Placebo	/	42 ± 11	100	10.01 ± 1.45
		40	SHR4640	2.5 mg	44 ± 12	100	10.05 ± 1.41
		40	SHR4640	5 mg	45 ± 14	100	9.77 ± 1.34
		40	SHR4640	10 mg	39 ± 13	97.5	10.34 ± 1.91
		39	Benzbromarone	50 mg	42 ± 12	100	9.91 ± 1.13

N, number of participants; *m*, mean; *SD*, standard deviation; *sUA*, serum uric acid

once daily: 3.37, 95% CI 1.81–6.26; RR of dotinurad 2 mg once daily vs. dotinurad 0.5 mg once daily: 3.48, 95% CI 1.93–6.28; RR of dotinurad 4 mg once daily vs. dotinurad 0.5 mg once daily: 3.99, 95% CI 2.24–7.11). Benzbromarone, dotinurad 2 mg once daily, and dotinurad 4 mg once daily showed better efficacy than probenecid (RR benzbromarone of vs. probenecid: 1.42, 95% CI 1.03–1.96; RR of dotinurad 2 mg once daily vs. probenecid: 1.47, 95% CI 1.01–2.14; RR of dotinurad 4 mg once daily vs. probenecid: 1.68, 95% CI 1.13–2.50). Dotinurad 4 mg once daily showed better efficacy than dotinurad 1 mg once daily, dotinurad 2 mg once daily, and SHR4640 (RR of dotinurad 4 mg once daily vs. dotinurad 1 mg once daily: 1.39, 95% CI 1.13–1.70; RR of dotinurad 4 mg once daily vs. dotinurad 2 mg once daily: 1.15, 95% CI 1.01–1.30; RR of dotinurad 4 mg once daily vs. SHR4640: 1.66, 95% CI 1.09–2.52).

Arhalofenate1	1.03 (0.48,2.19)	0.95 (0.46,1.96)	1.04 (0.50,2.20)	0.98 (0.48,2.01)	0.93 (0.42,2.04)	0.90 (0.44,1.86)	1.82 (0.58,5.75)	1.10 (0.54,2.27)	0.68 (0.33,1.41)	0.63 (0.28,1.38)	1.22 (0.56,2.67)	0.93 (0.46,1.87)
_	Dotinurad4	0.93 (0.65,1.31)	1.02 (0.85,1.23)	0.95 (0.71,1.28)	0.91 (0.63,1.30)	0.88 (0.65,1.20)	1.77 (0.69,4.56)	1.08 (0.77,1.51)	0.67 (0.41,1.08)	0.61 (0.36,1.03)	1.19 (0.83,1.70)	0.91 (0.68,1.22)
_	0.60 (0.03,13.06)	Verinurad	1.10 (0.80,1.51)	1.03 (0.79,1.34)	0.98 (0.65,1.47)	0.95 (0.74,1.23)	1.91 (0.75,4.87)	1.16 (0.90,1.50)	0.72 (0.47,1.10)	0.66 (0.41,1.06)	1.29 (0.86,1.92)	0.98 (0.81,1.19)
_	1.15 (1.01,1.30)	1.92 (0.09,42.19)	Dotinurad3	0.94 (0.73,1.20)	0.89 (0.64,1.25)	0.87 (0.66,1.13)	1.74 (0.69,4.42)	1.06 (0.78,1.43)	0.66 (0.41,1.04)	0.60 (0.36,0.99)	1.17 (0.84,1.63)	0.89 (0.69,1.15)
_	1.19 (0.94,1.49)	1.99 (0.09,43.64)	1.03 (0.86,1.25)	Benzbromarone	0.95 (0.65,1.38)	0.92 (0.80,1.07)	1.86 (0.76,4.56)	1.13 (0.88,1.44)	0.70 (0.46,1.07)	0.64 (0.40,1.02)	1.25 (0.86,1.80)	0.95 (0.80,1.14)
_	1.39 (1.13,1.70)	2.33 (0.11,51.33)	1.21 (0.97,1.52)	1.17 (0.87,1.57)	Dotinurad2	0.97 (0.66,1.42)	1.95 (0.74,5.16)	1.19 (0.80,1.77)	0.74 (0.44,1.24)	0.67 (0.38,1.18)	1.31 (0.90,1.91)	1.00 (0.70,1.43)
_	1.66 (1.09,2.52)	2.79 (0.13,61.70)	1.45 (0.97,2.16)	1.40 (0.99,1.99)	1.20 (0.76,1.90)	SHR4640	2.01 (0.81,4.99)	1.22 (0.96,1.55)	0.76 (0.50,1.15)	0.69 (0.44,1.11)	1.35 (0.93,1.96)	1.03 (0.87,1.22)
_	1.68 (1.13,2.50)	2.83 (0.13,63.07)	1.47 (1.01,2.14)	1.42 (1.03,1.96)	1.21 (0.78,1.88)	1.01 (0.63,1.63)	Probenecid	0.61 (0.24,1.54)	0.38 (0.14,1.01)	0.34 (0.13,0.95)	0.67 (0.25,1.77)	0.51 (0.21,1.28)
_	2.42 (0.34,17.50)	4.07 (0.18,90.38)	2.12 (0.29,15.25)	2.04 (0.28,14.75)	1.75 (0.24,12.72)	1.46 (0.20,10.66)	1.44 (0.19,10.66)	Lesinurad	0.62 (0.41,0.94)	0.57 (0.36,0.91)	1.11 (0.75,1.64)	0.84 (0.71,1.00)
_	4.82 (0.21,112.50)	8.09 (0.16,421.59)	4.20 (0.18,98.09)	4.06 (0.17,94.84)	3.47 (0.15,81.52)	2.90 (0.12,68.19)	2.86 (0.12,67.87)	1.99 (0.08,46.96)	Arhalofenate2	0.92 (0.59,1.41)	1.78 (1.06,3.00)	1.36 (0.93,1.99)
_	5.35 (0.23,126.52)	8.98 (0.17,472.96)	4.67 (0.20,110.31)	4.51 (0.19,106.66)	3.86 (0.16,91.67)	3.22 (0.14,76.68)	3.18 (0.13,76.32)	2.21 (0.09,52.81)	1.11 (0.41,2.99)	Arhalofenate3	1.95 (1.12,3.41)	1.49 (0.96,2.29)
_	3.99 (2.24,7.11)	6.70 (0.29,154.90)	3.48 (1.93,6.28)	3.37 (1.81,6.26)	2.88 (1.58,5.24)	2.40 (1.18,4.90)	2.37 (1.18,4.77)	1.65 (0.21,12.89)	0.83 (0.03,20.39)	0.75 (0.03,18.57)	Dotinurad1	0.76 (0.54,1.08)
_	38.79 (9.72,154.83)	65.14 (4.12,1029.52)	33.84 (8.49,134.90)	32.72 (8.21,130.45)	27.97 (6.93,112.89)	23.33 (5.75,94.75)	23.03 (5.56,95.31)	16.00 (3.90,65.61)	8.06 (0.47,136.62)	7.25 (0.42,124.62)	9.72 (2.17,43.44)	Placebo

Fig. 1 Risk ratios (RRs) with 95% confidence intervals (95% CIs) for primary outcomes. Data in light blue represent efficacy, and data in light green represent safety. The row treatment is compared with the column treatment. RR less than 1 favors the treatment in the row. Significant results are in red. Abbreviation: Dotinurad1, dotinurad 0.5 mg/day; Dotinurad2, dotinurad 1 mg/day; Dotinurad3, dotinurad 2 mg/day; Dotinurad4, dotinurad 4 mg/day; Arhalofenate1, arhalofenate 400 mg/day; Arhalofenate2, arhalofenate 600 mg/day; Arhalofenate3, arhalofenate 800 mg/day

We next estimated the efficacious rank of interventions by calculating the surface under the cumulative ranking (SUCRA), according to which the top two ranked interventions for achieving serum concentration of uric acid ≤ 6.0 mg/dL were dotinurad 4 mg once daily (SUCRA value 88.6) and verinurad (SUCRA value 78.2; Table 2). Figure 2A further shows the efficacy and safety curves of each treatment and also illustrated that dotinurad 4 mg once daily had the greatest area under the efficacy curve, which was in consistent with the following clustered ranking plot data (Fig. 2B).

Then, we conducted sensitivity analyses for primary efficacy and safety outcomes that excluded studies of high bias of risk and accounted for missing outcome data. The results showed that the effect estimates did not differ substantially (Supplementary Figure S4). Moreover, the missing outcome data also did not show substantial changes in SUCRA values (Table 2), further demonstrating the robustness of the model for the primary outcomes.

Primary safety outcome

Because of lacking unified definition regarding the safety outcome among the included studies, the primary safety outcome was defined as the incidence of any AEs in the present meta-analysis. Twelve studies [21–32] comprising 1833 patients reported the primary safety outcome. There were 13 individual nodes in the network plot for the incidence of

Table 2 SUCRA ranking of primary outcomes

Intervention	Efficacy		Safety		Intervention
	SUCRA	Missing data	SUCRA	Missing data	
Dotinurad4	88.6	88.9	90.1	89.7	Probenecid
Verinurad	78.2	78.0	81.9	79.2	Dotinurad1
Dotinurad3	75.2	75.8	72.8	73.3	Lesinurad
Benzbromarone	72.2	71.8	63.0	63.3	Dotinurad3
Dotinurad2	57.7	57.7	58.3	58.9	Dotinurad4
SHR4640	47.2	47.4	52.9	53.2	Arhalofenate1
Probenecid	46.2	46.6	51.9	51.9	Benzbromarone
Lesinurad	44.8	44.2	45.5	45.6	Verinurad
Arhalofenate2	34.3	34.2	42.0	42.5	Dotinurad2
Arhalofenate3	30.9	31.1	39.8	40.1	Placebo
Dotinurad1	23.2	23.0	34.1	34.1	SHR4640
Placebo	1.5	1.4	11.1	11.1	Arhalofenate2
/	/	/	6.8	7.1	Arhalofenate3

SUCRA, surface under the cumulative ranking; *Dotinurad1*, dotinurad 0.5 mg/day; *Dotinurad2*, dotinurad 1 mg/day; *Dotinurad3*, dotinurad 2 mg/day; *Dotinurad4*, dotinurad 4 mg/day; *Arhalofenate1*, arhalofenate 400 mg/day; *Arhalofenate2*, arhalofenate 600 mg/day; *Arhalofenate3*, arhalofenate 800 mg/day

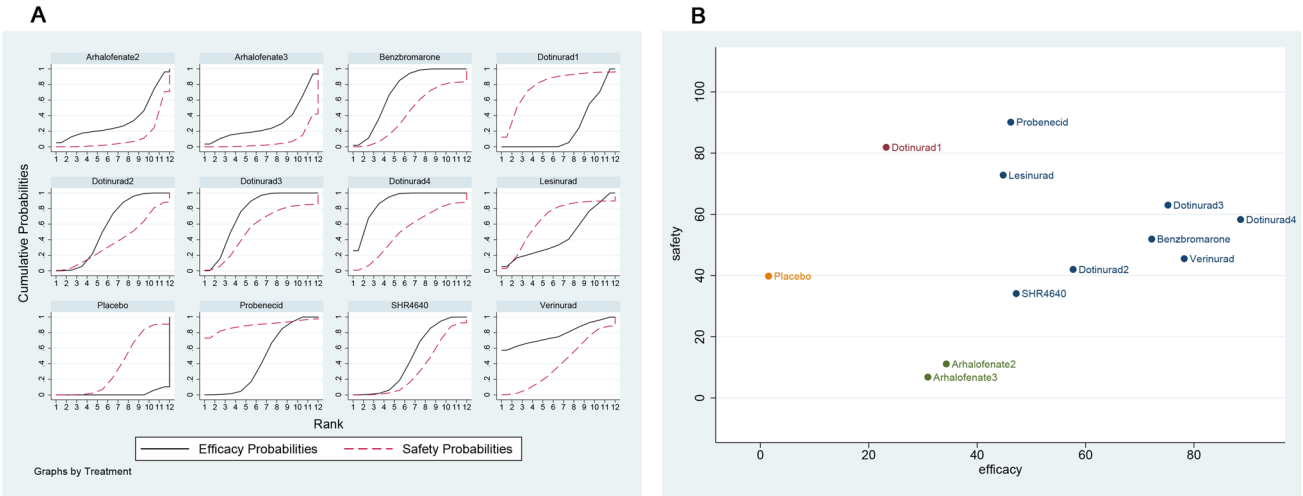

Fig. 2 **A** Cumulative ranking plots for primary outcomes. **B** Clustered ranking plot for primary outcomes. Abbreviation: Dotinurad1, dotinurad 0.5 mg/day; Dotinurad2, dotinurad 1 mg/day; Dotinurad3, dotinurad 2 mg/day; Dotinurad4, dotinurad 4 mg/day; Arhalofenate2, arhalofenate 600 mg/day; Arhalofenate3, arhalofenate 800 mg/day

AEs (Supplementary Figure S3). The network meta-analysis results showed that dotinurad 2 mg once daily, probenecid, and lesinurad had worse safety than arhalofenate 800 mg daily (RR of arhalofenate 800 mg daily vs. dotinurad 2 mg once daily: 0.60, 95% CI 0.36–0.99; RR of arhalofenate 800 mg daily vs. probenecid: 0.34, 95% CI 0.13–0.95; RR of arhalofenate 800 mg daily vs. lesinurad: 0.57, 95% CI 0.36–0.91; Fig. 1). The SUCRA values of interventions for primary safety outcome showed that the top two ranked interventions associated with more AE incidence were probenecid (SUCRA value 90.1) and dotinurad 0.5 mg once daily (SUCRA value 81.9; Table 2). Furthermore, cumulative ranking plot (Fig. 2A) reported that probenecid had the greatest area under the safety curve, and clustered ranking plot (Fig. 2B) also showed similar results.

Secondary outcomes

Four studies [22–25] comprising 801 patients reported the incidence of gouty arthritis. Seven studies [21, 26–31] comprising 1032 patients reported the incidence of gouty flares. The network plot for secondary outcomes included 6 individual nodes in gouty arthritis and 9 individual nodes in gouty flares (Supplementary Figure S3). The network meta-analyses showed no evidence of differences among all comparisons for gout arthritis and gout flares (Fig. 3A and 3B). The SUCRA values showed that the top two ranked interventions associated with increased incidence of gout arthritis were dotinurad 4 mg once daily (SUCRA value 90.1) and dotinurad 2 mg once daily (SUCRA value 60.3; Supplementary Table S2). Moreover, the top two ranked interventions associated with increased incidence of gout flares were arhalofenate 400 mg daily (SUCRA value 72.9) and SHR4640 (SUCRA value 64.9; Supplementary Table S2).

Certainty of evidence

The certainties of evidence for direct, indirect, and network estimates about primary outcomes are shown in Supplementary Table S3 and Supplementary Table S4. The network evidence for dotinurad 4 mg once daily was of moderate certainty compared with placebo. In the graded process, we downgraded the certainty of evidence to low or very low, mainly because of probable intransitivity, imprecision, and indirectness. The confidence of judgments in the estimates ranged from moderate to very low, indicating that further research is likely to influence our confidence in the estimate of effect. In addition, the comparison adjusted funnel plot did not show evidence of publication bias for small study effects (Supplementary Figure S5).

Heterogeneity and inconsistency

The heterogeneity of direct comparisons is displayed in Supplementary Table S5. Substantial heterogeneity was observed when comparing dotinurad 2 mg once daily with dotinurad 4 mg once daily for primary efficacy outcome ($I^2 = 52.40\%$) and when comparing dotinurad 2 mg once daily ($I^2 = 50.6\%$) or dotinurad 4 mg once daily ($I^2 = 71.2\%$) with placebo for primary safety outcome. However, there was no evidence showing heterogeneity in other comparisons for all outcomes. Furthermore, no local and global inconsistencies in all comparisons were observed (Supplementary Table S6 and Supplementary Table S7).

A

Dotinurad4	0.61 (0.19,1.95)	0.63 (0.35,1.13)	0.40 (0.12,1.38)	0.35 (0.06,2.00)	0.17 (0.02,1.32)
1.64 (0.51,5.22)	Dotinurad2	1.03 (0.32,3.28)	0.66 (0.14,3.24)	0.58 (0.09,3.78)	0.28 (0.03,2.40)
1.59 (0.89,2.85)	0.97 (0.30,3.10)	Dotinurad3	0.64 (0.22,1.90)	0.56 (0.10,3.18)	0.27 (0.03,2.11)
2.47 (0.72,8.45)	1.51 (0.31,7.37)	1.55 (0.53,4.58)	Benzbromarone	0.87 (0.11,6.73)	0.42 (0.04,4.27)
2.83 (0.50,16.01)	1.73 (0.26,11.29)	1.78 (0.31,10.06)	1.15 (0.15,8.84)	Dotinurad1	0.48 (0.04,6.11)
5.94 (0.76,46.76)	3.63 (0.42,31.62)	3.74 (0.47,29.39)	2.41 (0.23,24.71)	2.10 (0.16,26.94)	Placebo

B

Arhalofenate1	0.55 (0.11,2.73)	0.41 (0.11,1.60)	0.79 (0.22,2.78)	0.62 (0.16,2.32)	0.76 (0.29,1.99)	0.31 (0.02,4.66)	0.83 (0.25,2.82)	0.80 (0.25,2.51)
1.82 (0.37,9.01)	Arhalofenate2	0.75 (0.31,1.81)	1.44 (0.35,5.88)	1.12 (0.26,4.87)	1.38 (0.44,4.38)	0.56 (0.03,9.11)	1.52 (0.38,5.99)	1.45 (0.39,5.40)
2.43 (0.62,9.45)	1.34 (0.55,3.24)	Arhalofenate3	1.92 (0.58,6.36)	1.50 (0.42,5.32)	1.85 (0.76,4.46)	0.74 (0.05,11.01)	2.03 (0.64,6.44)	1.94 (0.66,5.72)
1.27 (0.36,4.45)	0.70 (0.17,2.85)	0.52 (0.16,1.73)	Benzbromarone	0.78 (0.23,2.64)	0.96 (0.43,2.16)	0.39 (0.03,4.34)	1.06 (0.51,2.21)	1.01 (0.36,2.80)
1.62 (0.43,6.08)	0.89 (0.21,3.86)	0.67 (0.19,2.36)	1.28 (0.38,4.31)	Lesinurad	1.23 (0.50,3.05)	0.50 (0.03,7.40)	1.35 (0.42,4.37)	1.29 (0.43,3.88)
1.32 (0.50,3.45)	0.72 (0.23,2.29)	0.54 (0.22,1.31)	1.04 (0.46,2.33)	0.81 (0.33,2.01)	Placebo	0.40 (0.03,5.14)	1.10 (0.52,2.31)	1.05 (0.57,1.95)
3.27 (0.21,49.82)	1.80 (0.11,29.47)	1.35 (0.09,19.96)	2.58 (0.23,28.93)	2.02 (0.14,30.17)	2.49 (0.19,31.76)	Probenecid	2.73 (0.22,34.09)	2.61 (0.19,35.95)
1.20 (0.35,4.05)	0.66 (0.17,2.60)	0.49 (0.16,1.57)	0.95 (0.45,1.98)	0.74 (0.23,2.39)	0.91 (0.43,1.92)	0.37 (0.03,4.58)	SHR4640	0.96 (0.36,2.52)
1.25 (0.40,3.93)	0.69 (0.19,2.56)	0.52 (0.17,1.52)	0.99 (0.36,2.74)	0.77 (0.26,2.32)	0.95 (0.51,1.77)	0.38 (0.03,5.27)	1.04 (0.40,2.75)	Verinurad

Fig. 3 Netleague plot for second outcomes. **A** Gout arthritis. **B** Gout flares. The row treatment is compared with the column treatment. RR less than 1 favors the treatment in the row. Significant results are in red. Abbreviation: Dotinurad1, dotinurad 0.5 mg/day; Dotinurad2, dotinurad 1 mg/day; Dotinurad3, dotinurad 2 mg/day; Dotinurad4, dotinurad 4 mg/day; Arhalofenate1, arhalofenate 400 mg/day; Arhalofenate2, arhalofenate 600 mg/day; Arhalofenate3, arhalofenate 800 mg/day. Significant results are in red

Discussion

To our knowledge, we conducted the first network meta-analysis investigating uricosuric agents for efficacy and safety in special gout or hyperuricemia patients. We analyzed 13 uricosuric interventions reported in 12 RCTs with 1851 participants. Overall, our results demonstrated that dotinurad 4 mg once daily provided higher efficacy and safety than other uricosuric agents in gout or hyperuricemia patients. As the present study was based on available uricosurics data, these findings may also be applied to special populations such as XOI intolerance or allergy patients and provide further comprehensive insight into the treatment value of certain uricosuric agents for gout or hyperuricemia that might be largely neglected previously.

According to a 2014 Cochrane review, uricosurics could be considered as second-line agents in the treatment of gout [33]. As a result, both the 2016 European League Against Rheumatism (EULAR) guideline and the 2020 American College of Rheumatology (ACR) guideline recommended to use uricosurics when gout or hyperuricemia patients were unable to tolerate or had contraindications to XOIs [34, 35]. However, these guidelines were mainly based on certain previous investigations that were limited by small sample sizes, uncontrolled trial designs, and lack of systematic clinical evidence regarding their benefit and safety [36–39]. As a matter of fact, considering that renal underexcretion of uric acid is the most frequent mechanism for the development of hyperuricemia and gout [10], uricosuric agents would have much wider application prospects in terms of their mechanism of action [40] and the potential treatment value of uricosurics might largely be neglected. In this regard, findings of the present network meta-analysis provided relatively large sample size and much more comprehensive evidence that may be used to inform further clinical decision-making in prevention and treatment of gout or hyperuricemia.

In the present study, we specifically evaluated each uricosuric agents to determine the most suitable treatment for gout or hyperuricemia, which suggested that dotinurad 4 mg once daily was more efficacious and safer than other uricosuric agents for achieving sUA target. Therefore, dotinurad can be considered as a promising therapeutic option in combination to XOIs if the treatment with XOI monotherapy fails to achieve target sUA. Furthermore, dotinurad might also be regarded as a potent uricosuric agent in the development of future evidence-based clinical practice guidelines for gout or hyperuricemia. This uricosuric treatment option will help to further improve the management of gout or hyperuricemia patients if used appropriately although further large-scale experiments and clinical studies are still needed to verify the present findings.

As for other uricosurics, the efficacy of verinurad was second only to dotinurad and its AE incidence was lower than dotinurad according to the present results, which might also be employed as an ideal uricosuric agent. Therefore, dotinurad and verinurad may be preferentially

recommended for patients with gout or hyperuricemia when planning to use uricosurics as therapeutic agents. However, although probenecid was the most widely used drug in gout or hyperuricemia patients in current clinical practice [41, 42], it did not show high efficiency in lowering sUA levels and ranked the worst in safety in the present study, which might mainly be ascribed to its wide interactions with other coadministered drugs [42]. In this regard, probenecid should be used cautiously with particular care especially for gout or hyperuricemia patients simultaneously administered with other drugs [43]. Additionally, benzbromarone showed moderate efficacy but also had relatively high risks of AEs, such as gout flares. As benzbromarone had broad application in clinic, further investigations regarding its clinical safety are still greatly warranted. In general, these findings provided advantageous evidence on the optimal option of uricosuric agents although more individualized treatment decisions should be made based on weight of the benefits against the AEs and costs of these therapy options to promote the comprehensive development of ULT.

Our findings have the following limitations. Firstly, the available data for many comparisons are based on few studies (in some cases just one or two studies), leading to relatively thinly connected networks and low statistical power to detect possible differences. Secondly, although we applied stringent inclusion criteria and identified no clear transitivity problems, there was heterogeneity for some outcomes. Therefore, the results should be interpreted with caution. Thirdly, due to limited original data, we were unable to perform subgroup analysis according to hyperuricemia type. Furthermore, most of the studies were of short duration (< 12 weeks) and the assessment of only short-term effects might overemphasize the therapeutic effects of uricosuric agents without considering special interest and serious AEs such as uric acid stone. Thus, the effects of long-term treatment require further investigation. Therefore, further studies, especially large-scale double-blind randomization trials, are required to provide more credible data for uricosuric agent treatment in gout or hyperuricemia.

Conclusion

The results of this network meta-analysis demonstrate an important absolute benefit of dotinurad 4 mg once daily to achieve target sUA and low risk of AEs for drug treatment of gout or hyperuricemia patients. Additionally, verinurad might be used as an alternative therapeutic option to dotinurad for second-line management of uricosuric agents. These findings provide novel potential treatment strategies that may improve the treatment efficacy in gout or hyperuricemia.

Abbreviations *sUA*: Serum uric acid; *ULT*: Urate-lowering therapy; *XOIs*: Xanthine oxidase inhibitors; *SUCRA*: Surface under the cumulative ranking; *GRADE*: Grade of Recommendations Assessment, Development, and Evaluation; *PRISMA*: Preferred Reporting Items for Systematic reviews and Meta-analyses

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Author contribution YJL, LRC, GHH, and SJY participate in the study design, conceptualization, and manuscript preparation. LRC, YJL, and ZLY are responsible for the literature search and screen. PW, FFJ, and YG are responsible for data acquisition. LRC, YJL, and MY are responsible for the quality assessment of included studies. LRC, YJL, and KQ are responsible for data analysis and interpretation of results. GHH and SJY are responsible for the manuscript editing and review. All authors reviewed the manuscript.

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Data availability The data that support the findings of this study and soft code are available in corresponding author.

Compliance with ethical standards

Disclosures None.

References

1. Singh JA, Gaffo A (2020) Gout epidemiology and comorbidities. Semin Arthritis Rheum 50(3S):S11–S16. https://doi.org/10.1016/j.semarthrit.2020.04.008
2. The RYM, Daniel K (2019) Inouye College of Pharmacy Scripts: perspectives on the epidemiology of gout and hyperuricemia. Hawaii J Med Public Health 78(2):71–76
3. Dehlin M, Jacobsson L, Roddy E (2020) Global epidemiology of gout: prevalence, incidence, treatment patterns and risk factors. Nat Rev Rheumatol 16(7):380–390. https://doi.org/10.1038/s41584-020-0441-1
4. Bove M, Cicero AF, Veronesi M, Borghi C (2017) An evidence-based review on urate-lowering treatments: implications for optimal treatment of chronic hyperuricemia. Vasc Health Risk Manag 8(13):23–28. https://doi.org/10.2147/VHRM.S115080
5. Pascart T, Richette P (2017) Current and future therapies for gout. Expert Opin Pharmacother 18(12):1201–1211. https://doi.org/10.1080/14656566.2017.1351945
6. Ying H, Yuan H, Tang X, Guo W, Jiang R, Jiang C (2021) Impact of serum uric acid lowering and contemporary uric acid-lowering therapies on cardiovascular outcomes: a systematic review and meta-analysis. Front Cardiovasc Med 23(8):641062. https://doi.org/10.3389/fcvm.2021.641062

7. Jansen TL, Tanja G, Matthijs J (2022) A historical journey of searching for uricosuric drugs. Clin Rheumatol 41(1):297–305. https://doi.org/10.1007/s10067-021-05930-1
8. Song D, Zhao X, Wang F, Wang G (2021) A brief review of urate transporter 1 (URAT1) inhibitors for the treatment of hyperuricemia and gout: current therapeutic options and potential applications. Eur J Pharmacol 15(907):174291. https://doi.org/10.1016/j.ejphar.2021.174291
9. Jones G, Panova E, Day R (2017) Guideline development for the management of gout: role of combination therapy with a focus on lesinurad. Drug Des Devel Ther 24(11):3077–3081. https://doi.org/10.2147/DDDT.S97959
10. Keenan RT (2017) Limitations of the current standards of care for treating gout and crystal deposition in the primary care setting: a review. Clin Ther 39(2):430–441. https://doi.org/10.1016/j.clinthera.2016.12.011
11. Shahid H, Singh JA (2015) Investigational drugs for hyperuricemia. Expert Opin Investig Drugs 24(8):1013–1030. https://doi.org/10.1517/13543784.2015.1051617
12. Perez-Ruiz F, Dalbeth N (2019) Combination urate-lowering therapy in the treatment of gout: what is the evidence? Semin Arthritis Rheum 48(4):658–668. https://doi.org/10.1016/j.semarthrit.2018.06.004
13. Borghi C, Perez-Ruiz F (2016) Urate lowering therapies in the treatment of gout: a systematic review and meta-analysis. Eur Rev Med Pharmacol Sci 20(5):983–992
14. Wu JY, Chang YT, Lin YC, Lee CH, Loh EW, Wu MY, Chang YS, Tam KW (2018) Efficacy and safety of lesinurad in patients with hyperuricemia associated with gout: a systematic review and meta-analysis of randomized controlled trials. Pharmacotherapy 38(11):1106–1119. https://doi.org/10.1002/phar.2183
15. Liu X, Zhai T, Ma R, Luo C, Wang H, Liu L (2018) Effects of uric acid-lowering therapy on the progression of chronic kidney disease: a systematic review and meta-analysis. Ren Fail 40(1):289–297. https://doi.org/10.1080/0886022X.2018.1456463
16. Zhang T, Pope JE (2017) Cardiovascular effects of urate-lowering therapies in patients with chronic gout: a systematic review and meta-analysis. Rheumatology (Oxford) 56(7):1144–1153. https://doi.org/10.1093/rheumatology/kex065
17. Kanji T, Gandhi M, Clase CM, Yang R (2015) Urate lowering therapy to improve renal outcomes in patients with chronic kidney disease: systematic review and meta-analysis. BMC Nephrol 19(16):58. https://doi.org/10.1186/s12882-015-0047-z
18. Takahashi T, Beppu T, Hidaka Y, Hosoya T (2021) Uric acid-lowering effect of dotinurad, a novel selective urate reabsorption inhibitor, in hypertensive patients with gout or asymptomatic hyperuricemia: a pooled analysis of individual participant data in phase II and III trials. Clin Exp Hypertens 43(8):730–741. https://doi.org/10.1080/10641963.2021.1950752
19. Blake KEG, Saag JL, Saag KG (2022) What's new on the frontline of gout pharmacotherapy? Expert Opin Pharmacother 23(4):453–464. https://doi.org/10.1080/14656566.2021.2020249
20. Hutton B, Salanti G, Caldwell DM et al (2015) The PRISMA extension statement for reporting of systematic reviews incorporating network meta-analyses of health care interventions: checklist and explanations. Ann Intern Med 162(11):777–784. https://doi.org/10.7326/M14-2385
21. Reinders MK, van Roon EN, Jansen TL, Delsing J, Griep EN, Hoekstra M, van de Laar MA, Brouwers JR (2009) Efficacy and tolerability of urate-lowering drugs in gout: a randomized controlled trial of benzbromarone versus probenecid after failure of allopurinol. Ann Rheum Dis 68(1):51–56. https://doi.org/10.1136/ard.2007.083071
22. Hosoya T, Sano T, Sasaki T, Fushimi M, Ohashi T (2020) Clinical efficacy and safety of dotinurad, a novel selective urate reabsorption inhibitor, in Japanese hyperuricemic patients with or without gout: randomized, multicenter, double-blind, placebo-controlled, parallel-group, confirmatory phase 2 study. Clin Exp Nephrol 24(Suppl 1):53–61. https://doi.org/10.1007/s10157-019-01818-2
23. Hosoya T, Sano T, Sasaki T, Fushimi M, Ohashi T (2020) Clinical efficacy and safety of dotinurad, a novel selective urate reabsorption inhibitor, in Japanese hyperuricemic patients with or without gout: an exploratory, randomized, multicenter, double-blind, placebo-controlled, parallel-group early phase 2 study. Clin Exp Nephrol 24(Suppl 1):44–52. https://doi.org/10.1007/s10157-019-01802-w
24. Hosoya T, Sano T, Sasaki T, Fushimi M, Ohashi T (2020) Dotinurad versus benzbromarone in Japanese hyperuricemic patient with or without gout: a randomized, double-blind, parallel-group, phase 3 study. Clin Exp Nephrol 24(Suppl 1):62–70. https://doi.org/10.1007/s10157-020-01849-0
25. Hosoya T, Fushimi M, Okui D, Sasaki T, Ohashi T (2020) Open-label study of long-term administration of dotinurad in Japanese hyperuricemic patients with or without gout. Clin Exp Nephrol 24(Suppl 1):80–91. https://doi.org/10.1007/s10157-019-01831-5
26. Tausche AK, Alten R, Dalbeth N, Kopicko J, Fung M, Adler S, Bhakta N, Storgard C, Baumgartner S, Saag K (2017) Lesinurad monotherapy in gout patients intolerant to a xanthine oxidase inhibitor: a 6 month phase 3 clinical trial and extension study. Rheumatology (Oxford) 1 56(12):2170–2178. https://doi.org/10.1093/rheumatology/kex350
27. NCT01399008 (2012) Safety/efficacy study to evaluate of MBX-102 in combination with allopurinol in gout patients. Available at: https://clinicaltrials.gov/ct2/show/NCT01399008
28. Fitz-Patrick D, Roberson K, Niwa K, Fujimura T, Mori K, Hall J, Yan X, Shen Z, Liu S, Ito Y, Baumgartner S (2019) Safety and efficacy of verinurad, a selective URAT1 inhibitor, for the treatment of patients with gout and/or asymptomatic hyperuricemia in the United States and Japan: findings from two phase II trials. Mod Rheumatol 29(6):1042–1052. https://doi.org/10.1080/14397595.2018.1538003
29. Shiramoto M, Liu S, Shen Z, Yan X, Yamamoto A, Gillen M, Ito Y, Hall J (2018) Verinurad combined with febuxostat in Japanese adults with gout or asymptomatic hyperuricaemia: a phase 2a, open-label study. Rheumatology (Oxford) 57(9):1602–1610. https://doi.org/10.1093/rheumatology/key100
30. Poiley J, Steinberg AS, Choi YJ, Davis CS, Martin RL, McWherter CA, Boudes PF, Arhalofenate Flare Study Investigators (2016) A randomized, double-blind, active- and placebo-controlled efficacy and safety study of arhalofenate for reducing flare in patients with gout. Arthritis Rheumatol 68(8):2027–2034. https://doi.org/10.1002/art.39684
31. Lin Y, Chen X, Ding H, Ye P, Gu J, Wang X, Jiang Z, Li D, Wang Z, Long W, Li Z, Jiang G, Li X, Bi L, Jiang L, Wu J, Guo L, Cai X, Lu X, Chen Q, Chen H, Peng A, Zuo X, Ning R, Zhang Z, Tai Y, Zhang T, Bao C (2021) Efficacy and safety of a selective URAT1 inhibitor SHR4640 in Chinese subjects with hyperuricaemia: a randomized controlled phase II study. Rheumatology (Oxford) 3 60(11):5089–5097. https://doi.org/10.1093/rheumatology/keab198
32. Ogino K, Kato M, Furuse Y, Kinugasa Y, Ishida K, Osaki S, Kinugawa T, Igawa O, Hisatome I, Shigemasa C, Anker SD, Doehner W (2010) Uric acid-lowering treatment with benzbromarone in patients with heart failure: a double-blind placebo-controlled crossover preliminary study. Circ Heart Fail 3(1):73–81. https://doi.org/10.1161/CIRCHEARTFAILURE.109.868604
33. Kydd AS, Seth R, Buchbinder R, Edwards CJ, Bombardier C (2014) Uricosuric medications for chronic gout. Cochrane Database Syst Rev 14(11):CD010457. https://doi.org/10.1002/14651858.CD010457.pub2
34. Richette P, Doherty M, Pascual E, Barskova V, Becce F, Castañeda-Sanabria J, Coyfish M, Guillo S, Jansen TL, Janssens

H, Lioté F, Mallen C, Nuki G, Perez-Ruiz F, Pimentao J, Punzi L, Pywell T, So A, Tausche AK, Uhlig T, Zavada J, Zhang W, Tubach F, Bardin T (2017) 2016 updated EULAR evidence-based recommendations for the management of gout. Ann Rheum Dis 76(1):29–42. https://doi.org/10.1136/annrheumdis-2016-209707

35. FitzGerald JD, Dalbeth N, Mikuls T, Brignardello-Petersen R, Guyatt G, Abeles AM, Gelber AC, Harrold LR, Khanna D, King C, Levy G, Libbey C, Mount D, Pillinger MH, Rosenthal A, Singh JA, Sims JE, Smith BJ, Wenger NS, Bae SS, Danve A, Khanna PP, Kim SC, Lenert A, Poon S, Qasim A, Sehra ST, Sharma TSK, Toprover M, Turgunbaev M, Zeng L, Zhang MA, Turner AS, Neogi T (2020) 2020 American College of Rheumatology Guideline for the Management of Gout. Arthritis Care Res (Hoboken) 72(6):744–760. https://doi.org/10.1002/acr.24180
36. Pui K, Gow PJ, Dalbeth N (2013) Efficacy and tolerability of probenecid as urate-lowering therapy in gout; clinical experience in high-prevalence population. J Rheumatol 40(6):872–876. https://doi.org/10.3899/jrheum.121301
37. Reinders MK, van Roon EN, Houtman PM, Brouwers JR, Jansen TL (2007) Biochemical effectiveness of allopurinol and allopurinol-probenecid in previously benzbromarone-treated gout patients. Clin Rheumatol 26(9):1459–1465. https://doi.org/10.1007/s10067-006-0528-3
38. Stocker SL, Graham GG, McLachlan AJ, Williams KM, Day RO (2011) Pharmacokinetic and pharmacodynamic interaction between allopurinol and probenecid in patients with gout. J Rheumatol 38(5):904–910. https://doi.org/10.3899/jrheum.101160
39. Perez-Ruiz F, Sundy JS, Miner JN, Cravets M, Storgard C, RDEA594-203 Study Group (2016) Lesinurad in combination with allopurinol: results of a phase 2, randomised, double-blind study in patients with gout with an inadequate response to allopurinol. Ann Rheum Dis 75(6):1074–1080. https://doi.org/10.1136/annrheumdis-2015-207919
40. Strilchuk L, Fogacci F, Cicero AF (2019) Safety and tolerability of available urate-lowering drugs: a critical review. Expert Opin Drug Saf 18(4):261–271. https://doi.org/10.1080/14740338.2019.1594771
41. Sun SS, Zhang DH, Shi Y, Lin CJ, Lin JY (2020) Efficacy and safety of urate-lowering treatments in patients with hyperuricemia: a comprehensive network meta-analysis of randomized controlled trials. J Clin Pharm Ther 45(4):729–742. https://doi.org/10.1111/jcpt.13156
42. Li S, Yang H, Guo Y, Wei F, Yang X, Li D, Li M, Xu W, Li W, Sun L, Gao Y, Wang Y (2016) Comparative efficacy and safety of urate-lowering therapy for the treatment of hyperuricemia: a systematic review and network meta-analysis. Sci Rep 8(6):33082. https://doi.org/10.1038/srep33082
43. Dubchak N, Falasca GF (2010) New and improved strategies for the treatment of gout. Int J Nephrol Renovasc Dis 3:145–166. https://doi.org/10.2147/IJNRD.S6048

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